



Lab 1B Terminology Glossary

This glossary covers every major term and concept used in Lab 1B. Use this as a quick reference while working through the lab.

AWS Services

EC2 (Elastic Compute Cloud)

What it is: Virtual servers in the cloud. Your "EC2 instance" is basically a computer running in AWS's data center that you control remotely.

In this lab: Your application runs on an EC2 instance that connects to RDS.

[!\[\]\(003082e50e3009141f59bd5df831749f_img.jpg\) AWS EC2 Documentation](#)

RDS (Relational Database Service)

What it is: Managed database service. AWS handles backups, patching, and maintenance—you just use the database.

In this lab: Your MySQL database runs on RDS. The EC2 app connects to it to store/retrieve data.

[!\[\]\(d3102649f02e825ddb76dc3de0190154_img.jpg\) AWS RDS Documentation](#)

SSM Parameter Store (Systems Manager Parameter Store)

What it is: A place to store configuration values (like settings, endpoints, feature flags). Think of it as a centralized config file in the cloud.

Best for:

- Non-sensitive configuration
- Values that don't need automatic rotation

- Application settings

In this lab: You store your DB endpoint, port, and database name here.

 [Parameter Store Documentation](#)

Secrets Manager

What it is: Secure storage specifically for sensitive data like passwords, API keys, and credentials. Supports automatic rotation.

Best for:

- Passwords and credentials
- Data that needs to be rotated automatically
- Sensitive API keys

In this lab: You store your database username and password here.

 [Secrets Manager Documentation](#)

CloudWatch

What it is: AWS's monitoring and observability service. It collects logs, tracks metrics, and triggers alarms.

Key components:

- **CloudWatch Logs** — Stores log files from your applications
- **CloudWatch Metrics** — Numerical data points over time
- **CloudWatch Alarms** — Notifications when metrics cross thresholds

In this lab: Your app sends error logs to CloudWatch, and an alarm fires when DB connection errors occur.

 [CloudWatch Documentation](#)

IAM (Identity and Access Management)

What it is: Controls *who* can do *what* in your AWS account. Manages users, roles, and permissions.

Key concepts:

- **IAM Role** — A set of permissions that can be assumed by AWS services (like EC2)
- **IAM Policy** — A document that defines specific permissions
- **Trust Relationship** — Defines which entities can assume a role

In this lab: Your EC2 instance has an IAM role that grants it permission to read from Parameter Store and Secrets Manager.

 [IAM Documentation](#)

Security Group

What it is: A virtual firewall that controls inbound and outbound traffic to AWS resources.

Think of it as: A bouncer at a door—it decides what network traffic is allowed in or out.

In this lab: The RDS Security Group must allow inbound traffic on port 3306 from your EC2's Security Group.

 [Security Groups Documentation](#)

VPC (Virtual Private Cloud)

What it is: Your own isolated network within AWS. Resources inside a VPC can communicate with each other, and you control how they connect to the outside world.

In this lab: Both your EC2 and RDS live within the same VPC so they can talk to each other.

 [VPC Documentation](#)

KMS (Key Management Service)

What it is: Manages encryption keys. When data is encrypted "at rest" in AWS, KMS often handles the keys.

In this lab: Secrets Manager uses KMS to encrypt your stored secrets. If a custom key is used, your EC2 role needs `kms:Decrypt` permission.

 [KMS Documentation](#)

Operational Concepts

Least Privilege

What it is: A security principle—give permissions for *only* what's needed, nothing more.

Example: Instead of giving EC2 access to *all* secrets, you only allow access to `lab/rds/mysql`.

Observability

What it is: The ability to understand what's happening inside a system by looking at its outputs (logs, metrics, traces).

The three pillars:

1. **Logs** — Detailed records of events
2. **Metrics** — Numerical measurements over time
3. **Traces** — Path of a request through systems

In this lab: CloudWatch Logs and Alarms give you observability into your application.

Incident Response

What it is: The process of detecting, diagnosing, and recovering from system failures.

In this lab: You simulate a failure, detect it via alarms, diagnose it via logs, and recover without redeploying.

MTTR (Mean Time to Recovery)

What it is: The average time it takes to restore service after an incident. Lower is better.

How this lab reduces MTTR:

- Stored configuration means you don't have to remember values
 - Centralized logs speed up diagnosis
 - Alarms notify you immediately
-

Secret Rotation

What it is: Automatically changing credentials on a schedule (e.g., every 30 days) to limit damage if they're compromised.

What breaks first: Usually the application—it has the old password while the database now expects the new one.

Metric Filter

What it is: A CloudWatch feature that scans log entries for specific patterns and converts matches into countable metrics.

In this lab: You create a filter that counts every log line containing "ERROR."

Technical Terms

ARN (Amazon Resource Name)

What it is: A unique identifier for any AWS resource.

Format:

```
arn:aws:service:region:account-id:resource
```

Example:

```
arn:aws:secretsmanager:us-east-2:919113286081:secret:lab/rds/
mysql-AbCdEf
```

JSON (JavaScript Object Notation)

What it is: A text format for structured data using key-value pairs.

Example:

```
{
  "username": "admin",
  "password": "secretpassword",
  "host": "mydb.rds.amazonaws.com"
}
```

SecureString

What it is: A Parameter Store parameter type that's encrypted using KMS.

Use for: Anything sensitive that you're storing in Parameter Store.

Log Group / Log Stream

- **Log Group:** A container for logs from a specific source (like `/aws/ec2/lab-rds-app`)
- **Log Stream:** Individual sequences of logs within a group (often one per instance)

Inline Policy

What it is: An IAM policy that's embedded directly in a role (as opposed to a standalone managed policy).

When to use: When the policy is specific to one role and won't be reused.

Where to Learn More

Free Resources

Resource	Best For
AWS Skill Builder	Free AWS training courses

Resource	Best For
AWS Well-Architected Labs	Hands-on exercises
Tutorials Dojo Cheat Sheets	Quick service overviews
AWS Documentation	Official deep-dive reference

Recommended Learning Path

1. **Understand IAM deeply** — Most AWS issues trace back to permissions
2. **Learn VPC networking** — Subnets, route tables, Security Groups
3. **Practice CloudWatch** — Logs, metrics, alarms, dashboards
4. **Study for AWS Solutions Architect Associate** — Ties everything together

YouTube Channels

- **Stephane Maarek** — Excellent AWS certification prep
- **Adrian Cantrill** — Deep technical explanations
- **Be A Better Dev** — Practical AWS tutorials